

1. Information about the indicator	
Goal	Conservation and rational use of oceans, seas and marine resources for sustainable development
Task	Minimize and eliminate the effects of ocean acidification, including through the development of scientific cooperation at all levels.
The indicator	The average value of the hydrogen index (pH) of seawater, measured in an agreed group of representative sampling stations (Caspian Sea)
2. Information about the organization	
Organization	RSE at the Kazhydromet Plant of the MENR RK
3. Definitions and concepts	
Definition	The average sea acidity is expressed as pH, the concentration of hydrogen ions on a logarithmic scale.
Basic concepts:	The indicator is calculated as the average numerical value for the year of the actual concentrations.
Justification and interpretation	It has been shown that the observed decrease in the pH of seawater affects a number of organisms and ecosystems, biodiversity and food security. This could negatively affect fishing and aquaculture, as well as other services, including tourism, transportation, and coastal protection.
4. Data sources and collection methods	
Data sources	The source is the «Information Bulletin on the State of the Environment of the Republic of Kazakhstan»
Data collection methods	Sampling of surface waters in accordance with the Instructional and Methodological Document "Organization and monitoring of surface water pollution of the Republic of Kazakhstan" (Appendix 3 to Order No. 624-O dated July 15, 2025), conducting laboratory analysis to determine the physico-chemical parameters of water, recording the analysis results in a journal and database, checking compliance of information quality with established standards, verification of the correctness of recording and conversion of observation data, calculations, verification of observational data for cases of deviations from established measurement and processing methods, as well as gross accidental errors (miscalculations) when performing measurements, preparation of generalized certificates with different time resolutions (monthly average, quarterly average, half-year average, annual average, etc.) and publication in the Newsletter on the state of the environment RK.
Unit of measurement	-
Periodicity	Every year
5. Calculation method and other methodological considerations	
Calculation method:	The indicator is calculated as the average numerical value for the year of the actual concentrations.
Comments and restrictions:	This indicator requires the collection of several observations in the form of separate data points to determine the variability of ocean acidity. According to this indicator, the average pH value of the Caspian Sea will be presented.
6. Data availability and disaggregation	
Data availability and gaps:	The data is available to the public on the official website of the Bureau of National Statistics of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan.

The level of disaggregation:	Countries provide complete datasets with relevant site-specific data and metadata files.
7. Comparability with international data/standards	Since this indicator only takes into account data provided only by Member States, there is no discrepancy between the estimates and the data sets provided.
8. References and documentation	https://stat.gov.kz/sustainable-development-goals/goal/11/